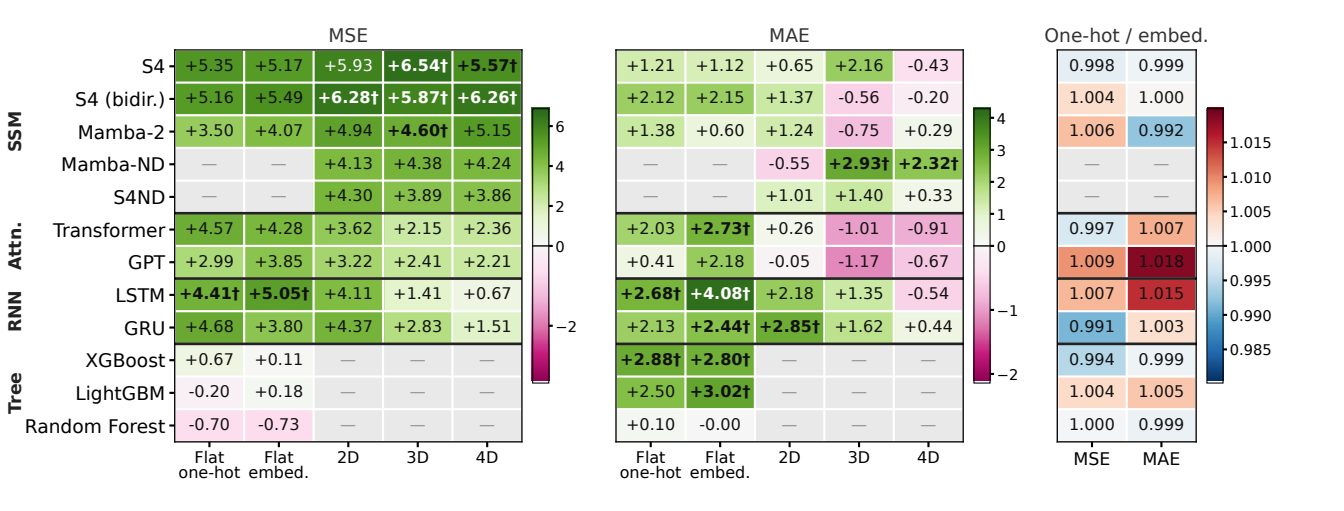
MSE
MAE
One-hot / embed.
SSM
Attn.
RNN
Tree
S4
S4 (bidir.)
Mamba-2
Mamba-ND
S4ND
Transformer
GPT
LSTM
GRU
XGBoost
LightGBM
Random Forest
+5.35 +5.17 +5.93 +6.54† +5.57†
+5.16 +5.49 +6.28† +5.87† +6.26†
+3.50 +4.07 +4.94 +4.60† +5.15
— — +4.13 +4.38 +4.24
— — +4.30 +3.89 +3.86
+4.57 +4.28 +3.62 +2.15 +2.36
+2.99 +3.85 +3.22 +2.41 +2.21
+4.41† +5.05† +4.11 +1.41 +0.67
+4.68 +3.80 +4.37 +2.83 +1.51
+0.67 +0.11 — — —
-0.20 +0.18 — — —
-0.70 -0.73 — — —
6
4
2
0
−2
Flat one-hot
Flat embed.
2D
3D
4D
+1.21 +1.12 +0.65 +2.16 -0.43
+2.12 +2.15 +1.37 -0.56 -0.20
+1.38 +0.60 +1.24 -0.75 +0.29
— — -0.55 +2.93† +2.32†
— — +1.01 +1.40 +0.33
+2.03 +2.73† +0.26 -1.01 -0.91
+0.41 +2.18 -0.05 -1.17 -0.67
+2.68† +4.08† +2.18 +1.35 -0.54
+2.13 +2.44† +2.85† +1.62 +0.44
+2.88† +2.80† — — —
+2.50 +3.02† — — —
+0.10 -0.00 — — —
4
3
2
1
0
−1
−2
Flat one-hot
Flat embed.
2D
3D
4D
0.998 0.999
1.004 1.000
1.006 0.992
— —
— —
0.997 1.007
1.009 1.018
1.007 1.015
0.991 1.003
0.994 0.999
1.004 1.005
1.000 0.999
1.015
1.010
1.005
1.000
0.995
0.990
0.985
MSE
MAE